exp262: does contacts-v1 need RoPE?

Issue [#262](https://github.com/Open-Athena/MarinFold/issues/262). contacts-v1 documents are self-describing and order-randomized — every statement carries its own `<pXXX>` coordinate and statement order is uniform noise — so the proposal is to delete RoPE and replace it with a width-3 causal *token smear*: mix the previous two tokens' embeddings into the current one, as in the nanogpt speedrun's smear module.

Phase 0 asks the trained default checkpoint whether there is anything to win, before any training is bought. Phase 1 is a 3-arm ablation at the exp232 1.5B architecture on a reduced token budget.

Two is the tight window

Enumerating the contacts-v1 grammar: a token's (statement form, slot) role is NOT a function of the previous 1 token class — `(POS, POS)` is ambiguous between a contact's second argument and a residue statement's head — and IS a deterministic function of the previous 2. Lookback 3 adds nothing. Width-3 smear is the tight bound.

What Phase 0 found

Three probes on `contacts-v1-exp199-cooldown-1.5B`, teacher-forced over ground-truth documents from the exp245 monomers, on a local A5000.

The smear half is directly motivated. Layer 1 holds two nearly pure previous-token heads (0.999 and 0.996 of their attention at offset 1) and a third splitting 0.75/0.16 across offsets 1 and 2 — a width-3 smear implemented in attention, at the cost of three heads.

The NoPE half loses its mechanism but keeps its premise. Co-referent retrieval is already distance-uniform out to 2048 tokens, so RoPE is not costing us long-range reach and the long-protein story is dead. But randomizing every exact cross-statement distance costs *less* than deterministically rescaling them at matched stretch, so the model genuinely is not reading those distances.

What position is actually for: counting. Keeping the position range fixed but letting two tokens share an id costs +1.23 nats, ten times any distance-randomizing intervention. The model uses position as an index, which is exactly what NoPE would take away — and exactly what the pre-registered stopping-behaviour guardrail was written to catch.

The pilot: the two changes only work together

15M-parameter twins on 150M tokens of the real decontaminated corpus, three seeds per arm at each arm's own best learning rate: smear alone buys 0.034 nats, NoPE alone COSTS 0.085, and together they buy 0.157 — an interaction of 0.208. The gain is entirely in the structure section, the shuffled bag where the theory said it should land. Note this overturns the Phase 0 reading, which had the smear as the safe half and NoPE as the speculative one.

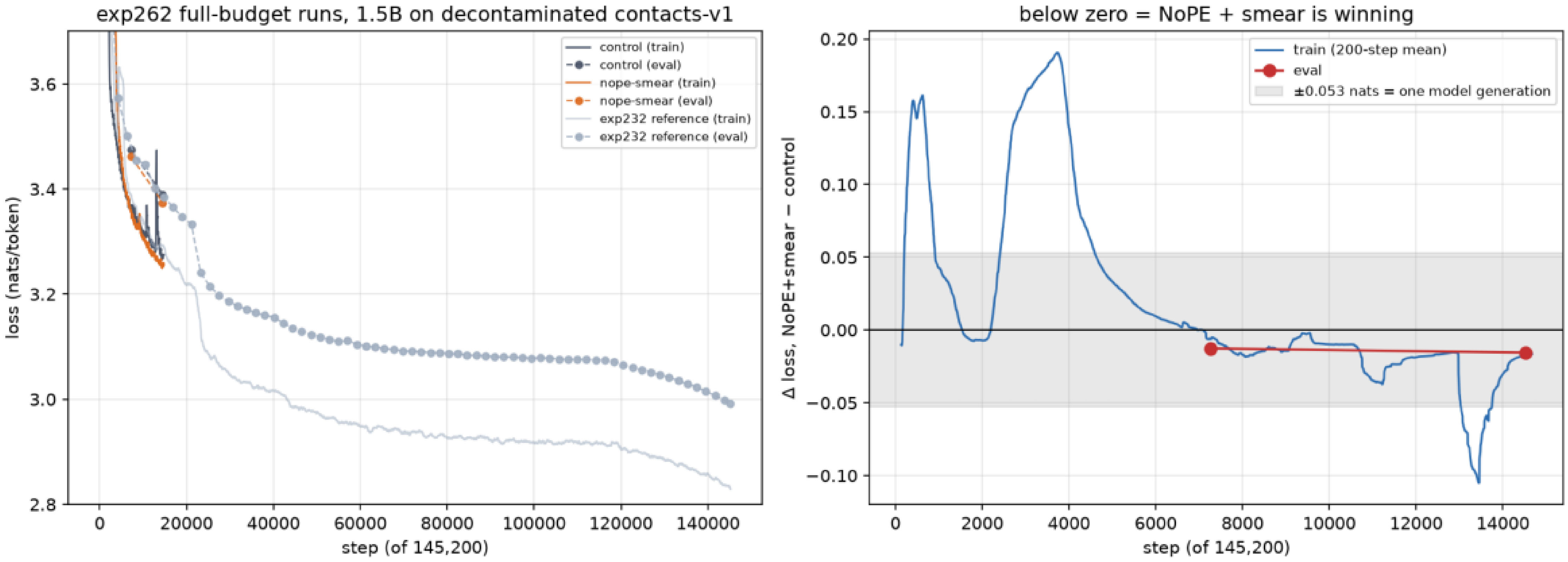
Where it stands

Two full-budget 1.5B runs are training on 64 H100s: exp232's usual setup against NoPE + smear, both at p06, differing in exactly one thing. At 10% of the schedule the new arm leads by 0.0156 nats on eval and the control is reproducing exp232 to 0.005 nats, which is what makes the gap readable.

Loss is not the deliverable. An accuracy claim needs a rollout R-precision eval, and that needs an HF exporter first — neither the NoPE config nor the smear weights have an HF Qwen3 representation.

full_run_progress.png

Left: both full-budget arms with exp232's run behind them in grey — the control is tracking it to ~0.01 nats, which is what licenses reading the gap at all. Right: the arm gap, against the 0.053 nats the whole #75 to #117 generation was worth. The early positive excursion is LR warmup; the crossover is near step 7,000.

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$ plot_full_runs.py
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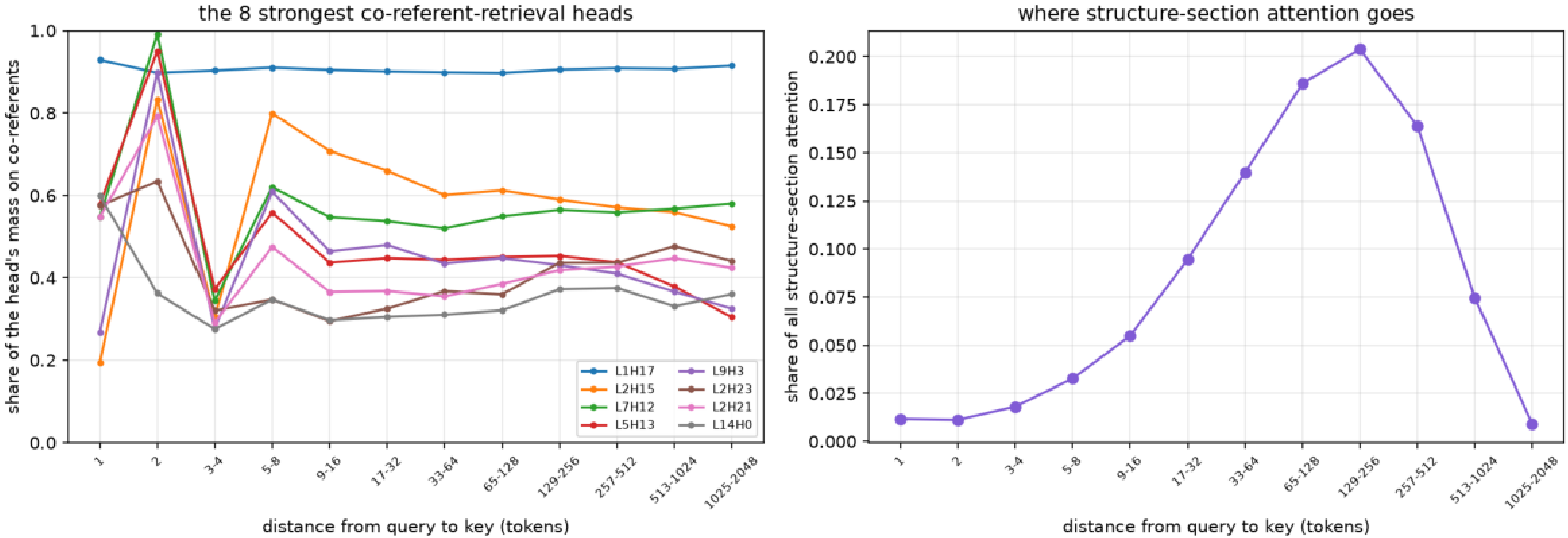
phase0_coreferent_retrieval.png

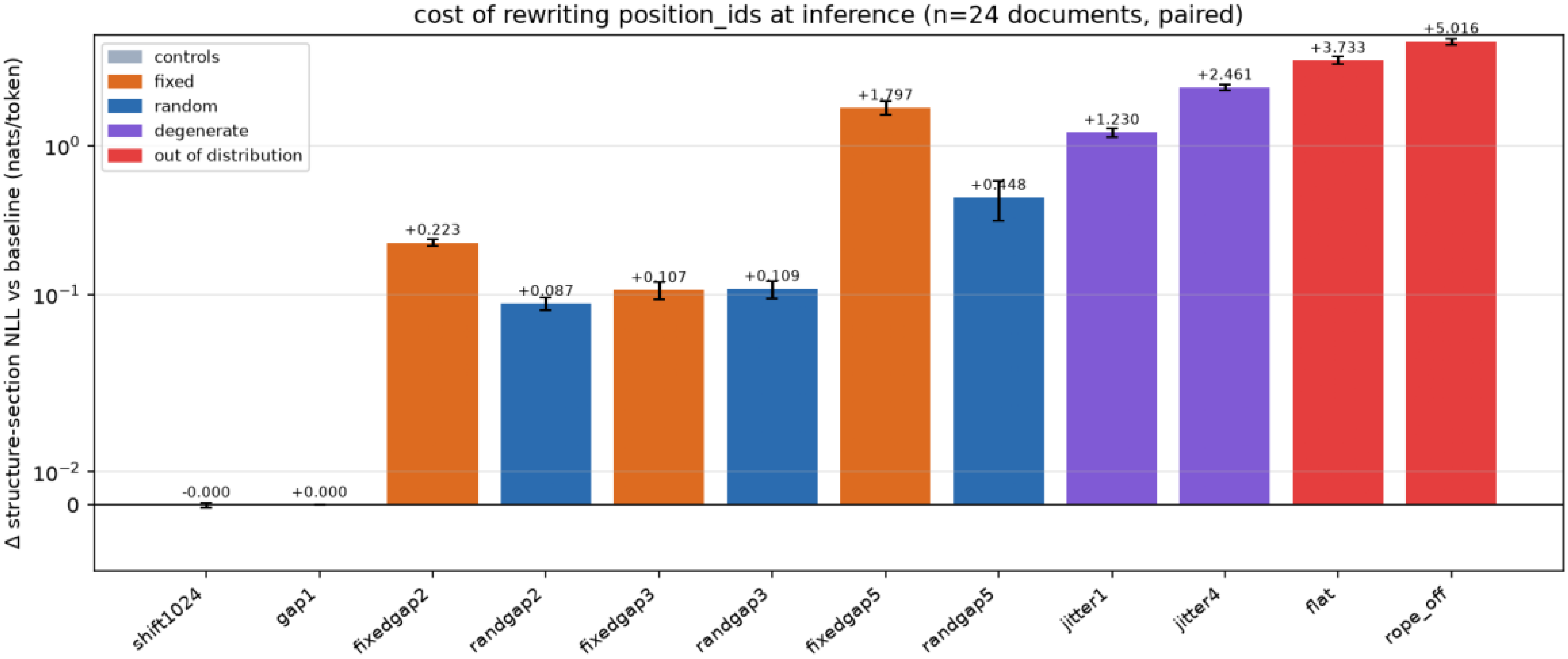
Left: for heads that retrieve earlier mentions of the query's own residue index, the share of their attention on those co-referents, by distance. Flat — L1H17 holds 0.90-0.93 out to 2048 tokens. Retrieval is already distance-uniform, so RoPE costs us no reach and the mechanistic case for dropping it fails. Right: where structure-section attention goes.



$ plot_phase0.py

phase0_position_interventions.png

Paired change in structure-section NLL when position_ids are rewritten (n=24). fixedgapN and randgapN match on expected stretch and never repeat a position; only randgapN destroys exact cross-statement distances, and is never the worse of the pair — the model does not read them. jitterN instead repeats ids at fixed range and costs 10x more: position is an index.

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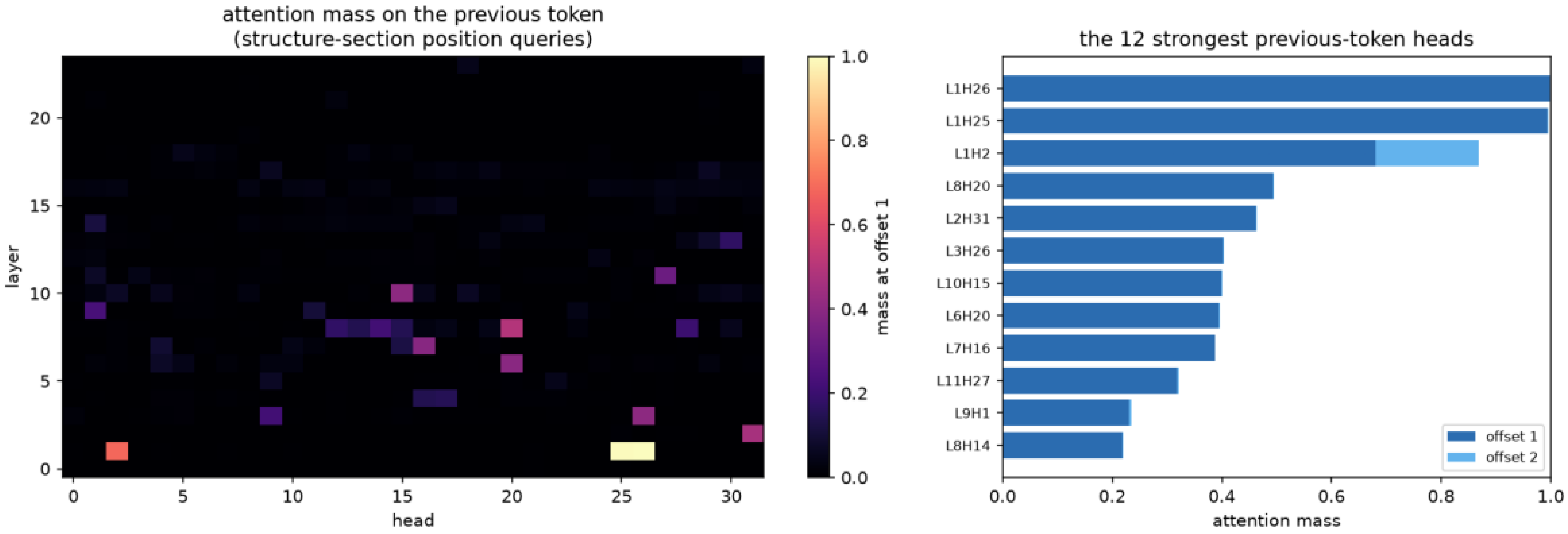
phase0_previous_token_heads.png

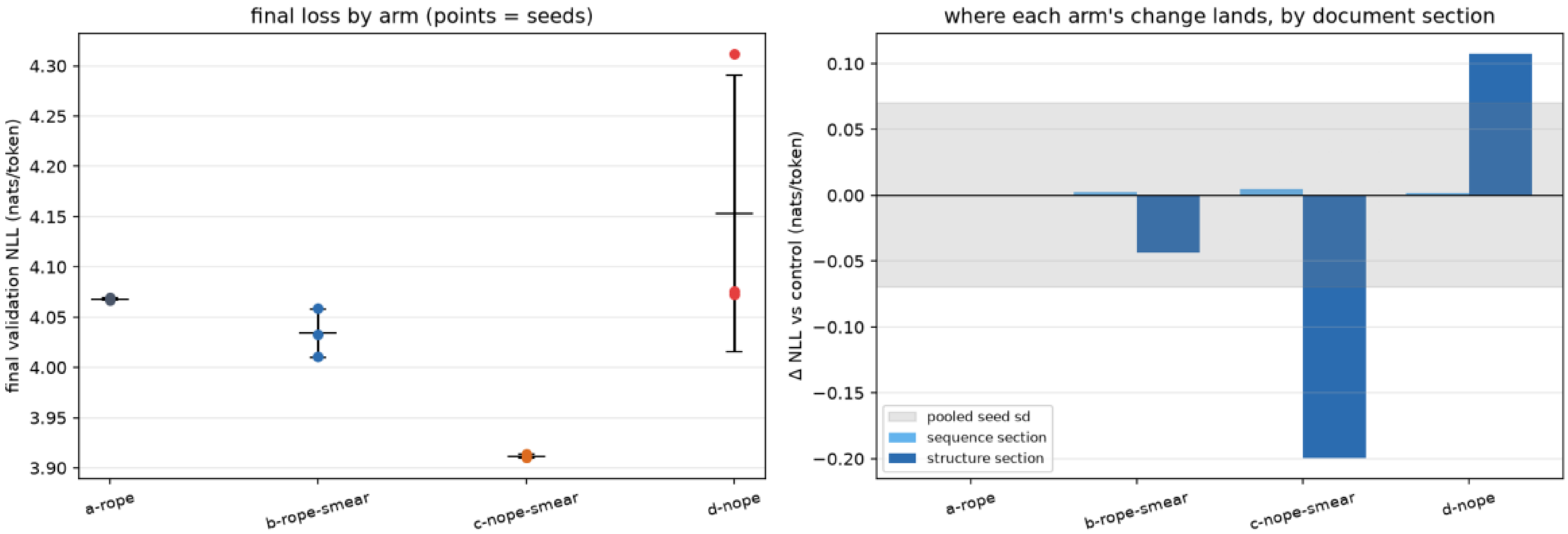
Attention mass at offset 1 per (layer, head), over structure-section position queries of 24 ground-truth documents. Layer 1 holds two near-pure previous-token heads (0.999, 0.996) and a third splitting 0.75/0.16 over offsets 1 and 2 — a width-3 smear built out of attention. Direct motivation for the smear half of #262.



$ plot_phase0.py

pilot_arm_comparison.png

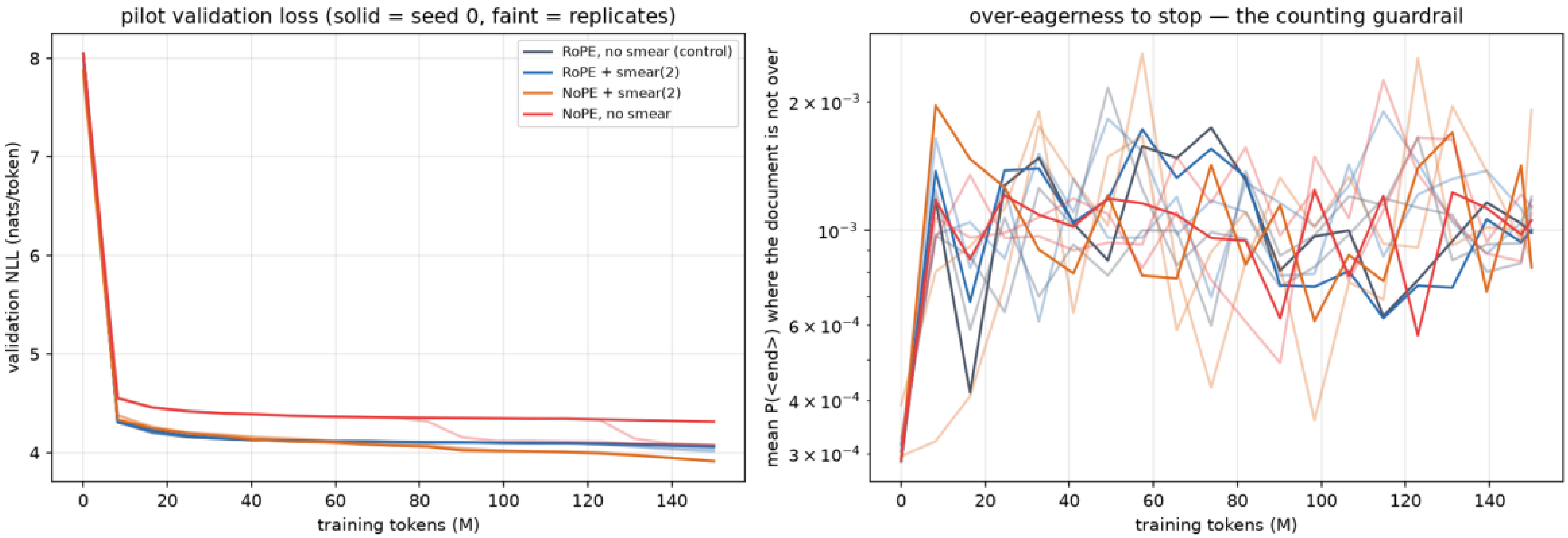
Left: final validation loss per arm, one point per seed, with the across-seed mean and spread. Right: each arm's gap to the control split by document section, against the pooled seed spread — the band is the pilot's resolution, and a bar inside it shows nothing.



$ analyze_pilot.py

pilot_loss_curves.png

Local pilot: 15M-parameter twins of the exp232 Qwen3 on real decontaminated contacts-v1, one line per seed. Right panel is the guardrail Phase 0 asked for — the probability the model puts on <end> when the document is not finished, which is the counting job NoPE takes away.



$ analyze_pilot.py